

Problem 1. Determine whether these statements are true or false.

- a) $\emptyset \in \{\emptyset\}$
- b) $\emptyset \in \{\emptyset, \{\emptyset\}\}$
- c) $\{\emptyset\} \in \{\emptyset\}$
- d) $\{\emptyset\} \in \{\{\emptyset\}\}$
- e) $\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$
- f) $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$
- g) $\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\}$

Answer.

- a) True
- b) True
- c) False
- d) True
- e) True
- f) True
- g) True

Problem 2. Find two sets A and B such that $A \in B$ and $A \subseteq B$.

Answer. $A = \emptyset, B = \{\emptyset\}$

Problem 3. Show that if $A \subseteq C$ and $B \subseteq D$, then $A \times B \subseteq C \times D$.

Answer. since $x \in A \wedge x \in C$ and $y \in B \wedge y \in D$ from the definition of subsets, $(x, y) \in A \times B$ and $(x, y) \in C \times D$. C and D is a larger set that contains A and D , therefore the product of A and B also must be in C and D .

Problem 4. This exercise presents **Russell's paradox**. Let S be the set that contains a set x if the set x does not belong to itself, so that $S = \{x \mid x \notin x\}$.

- a) Show the assumption that S is a member of S leads to a contradiction.
- b) Show the assumption that S is not a member of S leads to a contradiction.

By parts (a) and (b) it follows that the set S cannot be defined as it was. This paradox can be avoided by restricting the types of elements that sets can have.

Answer.

- a) If S belongs to itself, then it contradicts the definition where the set should not contain any set x that contains itself.
- b) If S does not belong to itself, then it contradicts the definition where it should contain itself; since x is any set that does not contain itself.

Problem 5. Prove the idempotent laws in Table 1 by showing that

- a) $A \cup A = A$
- b) $A \cap A = A$

Answer.

- a) $A \cup A = \{x \mid x \in (A \cup A)\}$
 $= \{x \mid x \in A \vee x \in A\}$ (def of union)
 $= \{x \mid x \in A\}$ (simplify)
 $= A$
- b) $A \cap A = \{x \mid x \in (A \cap A)\}$ (def of intersection)
 $= \{x \mid x \in A \wedge x \in A\}$ (def of intersection)
 $= \{x \mid x \in A\}$ (simplify)
 $= A$

Problem 6. Let A and B be sets. Show that

- a) $(A \cap B) \subseteq A$
- b) $A \subseteq (A \cup B)$
- c) $A - B \subseteq A$
- d) $A \cap (B - A) = \emptyset$
- e) $A \cup (B - A) = A \cup B$

Answer.

- a) by definition of a subset, we want to prove that $\forall x((x \in (A \cap B)) \rightarrow (x \in A))$
 - let x be an arbitrary element so that $x \in (A \cap B)$
 - by definition of intersection, $(x \in A) \wedge (x \in B)$
 - therefore, $x \in A$ must be true, completing the implication. ■
- b) by definition of a subset, we want to prove $\forall x((x \in A) \rightarrow (x \in (A \cup B)))$
 - let x be an arbitrary element so that $x \in A$.
 - by rule of addition, since $x \in A$ is true, $(x \in A) \vee (x \in B)$ is also true.
 - by definition of a union, $x \in (A \cup B)$
 - therefore, we proved $(x \in A) \rightarrow (x \in (A \cup B))$, satisfying the definition of a subset. ■
- c) by definition of a subset, we want to prove $\forall x((x \in (A - B)) \rightarrow (x \in A))$.
 - let x be an arbitrary element so that $x \in (A - B)$.
 - by definition of a set difference, $(x \in A) \wedge (x \notin B)$.
 - by simplification, we can get $x \in A$, which completes our implication. ■
- d) let x be an arbitrary element in $A \cap (B - A)$.
 - by definition of intersection, $(x \in A) \wedge (x \in B - A)$.
 - by definition of set difference, $(x \in A) \wedge (x \in B) \wedge (x \notin A)$.
 - x cannot be both in A and not in A , therefore, we get an empty set. ■
- e) let x be an arbitrary element in $A \cup (B - A)$.
 - by definition of union, $(x \in A) \vee (x \in B - A)$.
 - by definition of set difference, $(x \in A) \vee ((x \in B) \wedge (x \notin A))$.
 - by distribution, we get $((x \in A) \vee (x \in B)) \wedge ((x \in A) \vee (x \notin A))$
 - since $(x \in A) \vee (x \notin A)$ is a tautology, so we can discard the right part of the $\vee$ operator.
 - after simplifying, we get $(x \in A) \vee (x \in B)$, and by definition of a union, we know that $(x \in A \cup B)$. ■

Problem 7. Show that if A and B are sets with $A \subseteq B$, then

- a) $A \cup B = B$
- b) $A \cap B = A$

Answer.

- a) let x be an arbitrary element in $A \cup B$.
 - by definition of union, $(x \in A) \vee (x \in B)$.
 - if $(x \in A)$, then since $\forall x((x \in A) \rightarrow (x \in B))$, x is in B .
 - if $(x \in B)$, that satisfies our case.
 - both cases show that for any x , $x \in B$. therefore, we can say that $A \cup B = B$. ■
- b) let x be an arbitrary element in $A \cap B$.
 - by definition of intersection, $(x \in A) \wedge (x \in B)$.
 - since both side of the $\wedge$ operator are true, we can simplify and get $(x \in A)$. ■

Problem 8. Let A , B , and C be sets. Use the identity $A - B = A \cap \bar{B}$, which holds for any sets A and B , and the identities from Table 1 to show that $(A - B) \cap (B - C) \cap (A - C) = \emptyset$.

Answer.

$$\begin{aligned} & (A - B) \cap (B - C) \cap (A - C) \\ &= (A \cap \bar{B}) \cap (B \cap \bar{C}) \cap (A \cap \bar{C}) && \text{(substitute using identity)} \\ &= (A \cap A) \cap (B \cap \bar{B}) \cap (\bar{C} \cap \bar{C}) && \text{(rearrange intersections)} \\ &= A \cap \emptyset \cap \bar{C} && \text{(simplify)} \\ &= \emptyset && \text{(anything intersect with } \emptyset \text{ is } \emptyset) \end{aligned}$$

Challenge 1. The **symmetric difference** of A and B , denoted by $A \oplus B$, is the set containing those elements in either A or B , but not in both A and B .

Determine whether the symmetric difference is associative; that is, if A , B , and C are sets, does it follow that $A \oplus (B \oplus C) = (A \oplus B) \oplus C$?

Answer. By the definition of symmetric difference, $A \oplus B = (A \cup B) - (A \cap B)$. Therefore,

$$A \oplus (B \oplus C) = (A \cup ((B \cup C) - (B \cap C))) - (A \cap ((B \cup C) - (B \cap C)))$$

$$(A \oplus B) \oplus C = (((A \cup B) - (A \cap B)) \cup C) - (((A \cup B) - (A \cap B)) \cap C)$$